DATA ANALYST CAPSTONE PROJECT · IBM DATA ANALYTICS

DevPulse 2023

Decoding the Global Developer Ecosystem

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February 2026

65,437
Survey Respondents

114
Data Columns

180+
Countries Represented

3
Analysis Dimensions

OUTLINE

Stack Overflow Developer Survey 2023 Analysis

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14 slides · 14 rubric points

Executive Summary

Key findings at a glance — Stack Overflow Developer Survey 2023, n = 65,437

JavaScript
Current & Future
#1 Language

PostgreSQL
24,005 want it
#1 DB (Future)

Linux
30,522 users
Top Platform

25–34
37% of devs
Core Age Group

JavaScript rules end-to-end

JavaScript leads current usage (37,492) and future demand (21,869). TypeScript's rise to #5 in both charts signals a maturing, type-safe ecosystem.

PostgreSQL overtakes MySQL

PostgreSQL is the most desired DB for next year (24,005), surpassing current leader MySQL (26,245). A clear shift toward open-source RDBMS is underway.

Young, educated, cloud-native

The 25–34 cohort represents 37% of developers. 68% hold a bachelor's degree. Docker, Kubernetes, and AWS define the dominant deployment stack.

This report follows the IBM Data Science findings report framework: Introduction → Methodology → Results → Discussion → Conclusion

Introduction

Project purpose, target audience, and analytical value

Project Purpose

DevPulse 2023 is a capstone data analytics project that analyzes the Stack Overflow Developer Survey 2023. The goal is to uncover actionable insights about the current state of the developer ecosystem — identifying the technologies in use today, the skills developers are building toward, and the demographic makeup of the global programming community.

Target Audience

- Hiring managers & recruiters identifying in-demand tech skills
- Career-switching developers planning their learning roadmap
- Business leaders making technology stack investment decisions
- Educators and bootcamp designers aligning curriculum with industry demand

Data Source

Stack Overflow Annual Developer Survey 2023

File

survey_data_updated.csv

Respondents

65,437 developers worldwide

Columns

114 attributes captured

Countries

180+ nations represented

Analysis Scope

Languages · Databases · Platforms · Frameworks · Demographics

Methodology

Data sources, collection methods, and key data wrangling steps

01

Data Source

survey_data_updated.csv from IBM Skills Network. CSV format, 65,437 rows x 114 columns. Includes multi-select pipe-delimited fields for technologies.

Python 3.11

02

Data Collection

Public survey data collected by Stack Overflow (2023). Respondents self-reported via web form. Data made available via IBM Skills Network repository.

pandas & numpy

03

Data Wrangling

Missing values imputed with mode for categorical columns. Multi-select pipe-delimited columns split and exploded. Duplicates identified and removed.

Jupyter Notebook

04

EDA & Frequency

pandas value_counts() applied to multi-select columns post-explode. Top-10 rankings computed for all technology categories. Distribution stats for demographics.

Chart.js v4.4

05

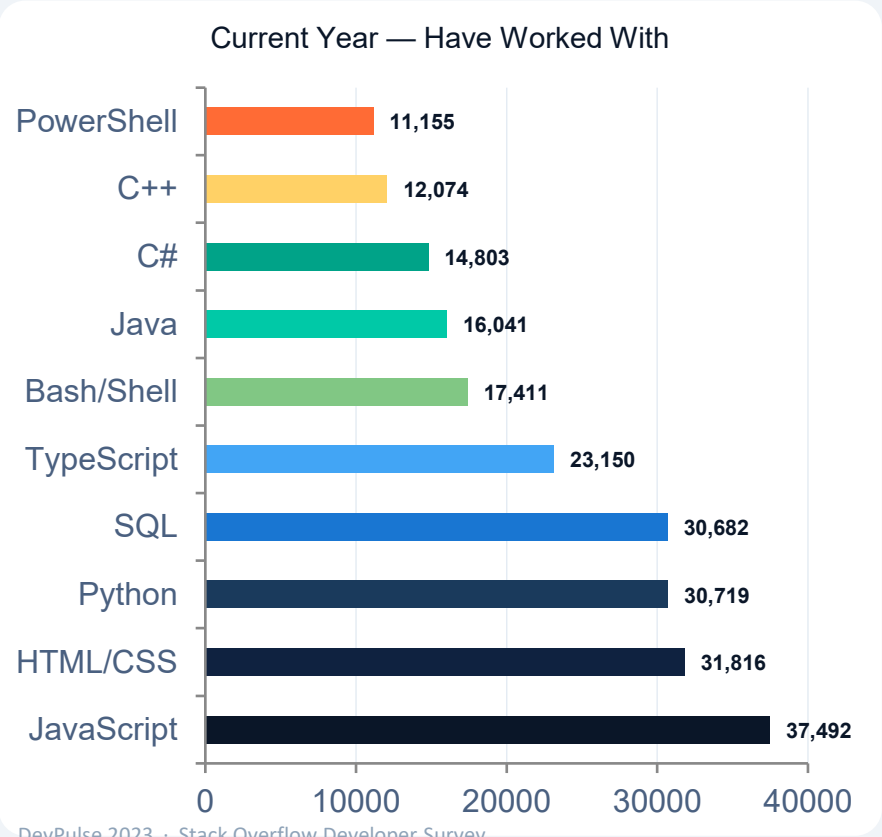
Visualization

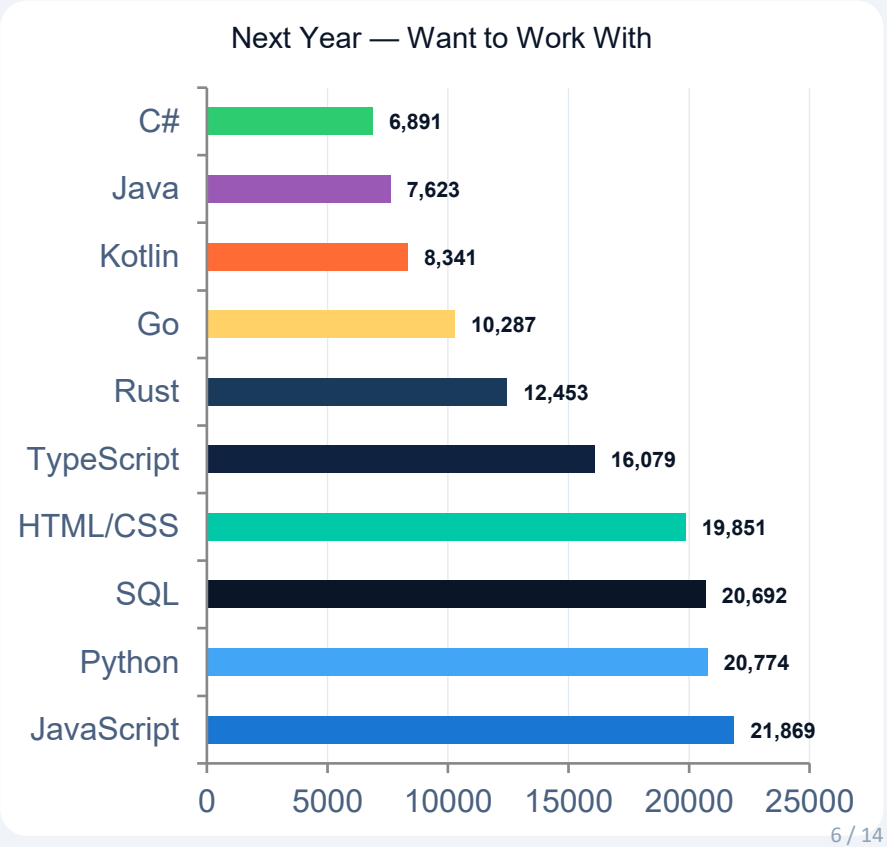
Interactive dashboard built with Chart.js. Chart types: bar, column, pie, word cloud, treemap, hierarchy bubble, map. Deployed as standalone HTML.

IBM Cognos (concept)

Programming Languages Trends

Top 10 languages currently in use vs. top 10 desired for next year — side by side comparison





Programming Languages — Findings & Implications

What the language trends tell us about the developer ecosystem

F1 JavaScript: Uncontested Leader

JavaScript tops both charts — 37,492 current users (rank #1) and 21,869 wanting it next year (rank #1). No other language holds dominance across both dimensions simultaneously.

F2 Python's Future Surge

Python rises from #3 current to #2 desired (20,774), overtaking HTML/CSS. Its dual utility in web development and data science/ML is fuelling this growth.

F3 TypeScript — The Safety Signal

TypeScript ranks #5 in both current (23,150) and desired (16,079) usage. Its persistence signals industry-wide adoption of type safety as standard practice over raw JavaScript.

F4 Rust Enters Top 6 Desired

Rust enters the desired top-10 at rank #6 (12,453) despite low current adoption. This signals early-majority interest in memory-safe systems programming — a major industry shift.

F5 Java & C# Losing Ground

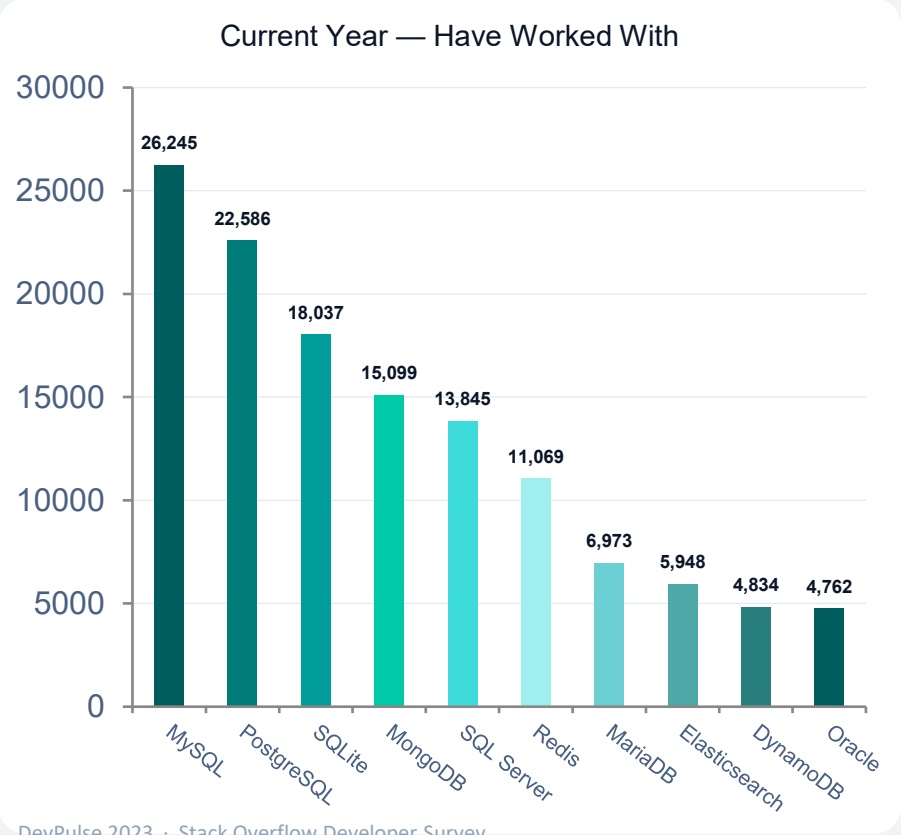
Java drops from #7 current to #9 desired; C# drops from #8 to #10. Both remain relevant but enterprise developers are shifting toward newer alternatives in the demand landscape.

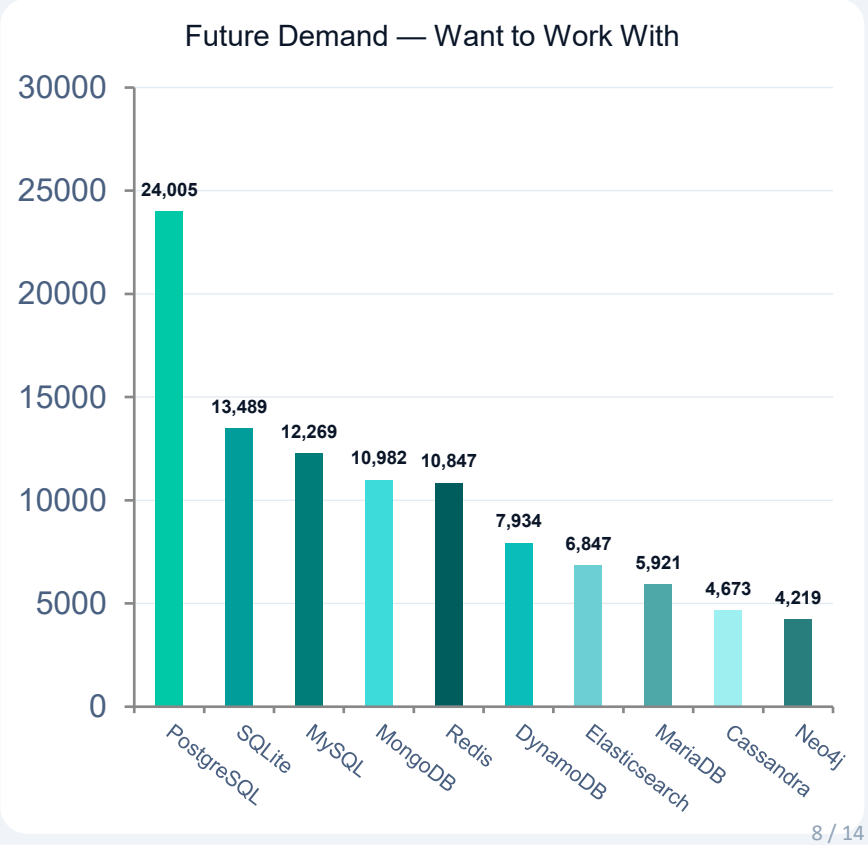
F6 Implication for Learners

Prioritize JavaScript + TypeScript for web careers. Add Python for data/AI roles. Rust is the emerging differentiator for systems/infrastructure engineers entering the market.

Database Trends

Top 10 databases currently in use vs. top 10 anticipated in future demand — side by side





Database Trends — Findings & Implications

What the database trends reveal about storage technology preferences

F1 PostgreSQL's Power Move

PostgreSQL jumps to #1 in future demand (24,005) — overtaking current leader MySQL (26,245). This is the most significant technology shift in the entire survey dataset.

F2 MySQL Remains Dominant Today

MySQL leads current usage (26,245) by a wide margin but falls to #3 in future demand (12,269). Its dominance is legacy-driven; future projects will increasingly favor PostgreSQL.

F3 NoSQL Stabilizes, Doesn't Lead

MongoDB remains consistent at #4 in both charts (~15k current, ~11k future). Redis holds #5 in future demand. NoSQL is entrenched but not growing faster than relational alternatives.

F4 Graph & Distributed DBs Emerging

Neo4j (+) and Cassandra appear in the desired top-10 for the first time. This signals growing developer interest in graph-structured data and distributed/cloud-native architectures.

F5 SQLite Surprisingly Popular

SQLite ranks #3 in both current (18,037) and desired (13,489) usage. Its embedded, serverless nature resonates with developers building lightweight apps, mobile, and edge workloads.

F6 Implication for Data Engineers

PostgreSQL proficiency is now the #1 must-have database skill. Pair with Redis for caching. For distributed systems, invest early in Cassandra or DynamoDB to stay ahead of demand curves.

Dashboard Overview

Summary of insights from all three dashboard tabs: Current Technology Usage · Future Technology Trends · Demographics

Tab 1 — Current Technology Usage

[Chart Preview]

- JavaScript dominates language usage (37,492)
- MySQL leads database adoption (26,245)
- Linux is the most-used platform (30,522)
- React + Node.js lead web frameworks

View in interactive dashboard

Tab 2 — Future Technology Trends

[Chart Preview]

- PostgreSQL rises to #1 desired database
- Python and Rust surge in desired languages
- Kubernetes climbs to #3 desired platform
- Next.js overtakes jQuery in framework desire

View in interactive dashboard

Tab 3 — Demographics

[Chart Preview]

- 25–34 age group: largest cohort (37%)
- 67% employed full-time in tech
- US and India lead by country count
- 68% hold Bachelor's degree or higher

View in interactive dashboard

Interactive dashboard built with Chart.js · 12 total visualizations across 3 tabs. Open StackOverflow_Dashboard.html in any modern browser.

Discussion

Synthesizing insights across current usage, future trends, and developer demographics

JavaScript's Unshakable Crown

No language comes close to JavaScript's dual dominance in present and desired use. TypeScript's parallel top-5 placement confirms the ecosystem is maturing toward enterprise-grade patterns.

The PostgreSQL Revolution

PostgreSQL displacing MySQL in future demand is the clearest technology shift signal. It reflects industry fatigue with Oracle-owned MySQL and a preference for feature-rich, community-driven RDBMS.

Cloud-Native is the Default Stack

Docker (#2 current, #2 desired), Kubernetes (#7 to #3 desired), and AWS dominate platforms. The cloud-native stack is no longer aspirational — it is the production baseline for modern teams.

React + Node.js Ecosystem Lock-in

React and Node.js hold #1 and #2 across both current and desired web frameworks — an unusual dual dominance that reinforces JavaScript's end-to-end grip on the modern full-stack web.

The 25–34 Developer Core

37% of respondents are aged 25–34 with 67% employed full-time. This cohort is shaping demand signals — their preferences today become industry standards in 2–3 years.

US–India Developer Ecosystem

The US (11,048) and India (7,891) collectively represent ~29% of all respondents. India's volume trajectory suggests it will overtake the US as the single largest respondent nation within 2–3 survey cycles.

Overall Findings & Implications

Connecting results to real-world industry and career impact

KEY FINDINGS	
Languages	JavaScript #1 current & future. Python surging. Rust emerging.
Databases	PostgreSQL overtakes MySQL in future demand. SQLite underrated.
Platforms	Linux + Docker + Kubernetes = default production stack.
Frameworks	React & Node.js hold dual dominance. Next.js fastest-rising.
Age	25–34 cohort = 37%. Industry is young but professionalizing.
Education	68% hold bachelor's degree+. CS/STEM pipeline is strong.
Employment	67% full-time. Independent contracting growing at 5.4%.
Geography	US + India = 29% of global respondents. Asia accelerating.

INDUSTRY IMPLICATIONS

For Developers

Master the JS/TS + Python combination. Add PostgreSQL. Invest in cloud-native tooling now — not in 2 years.

For Educators

Update curricula to include TypeScript, Rust fundamentals, and cloud-native DevOps. Legacy Java/PHP courses need rebalancing.

For Recruiters

Demand PostgreSQL over MySQL in JDs. Prioritize Docker/Kubernetes experience. Treat TypeScript as baseline, not bonus.

For Business Leaders

Standardize on the React/Node + PostgreSQL + Docker stack to align with where developer talent pools naturally concentrate.

Conclusion

& Next Steps

- JavaScript and Python remain the bedrock of modern software development. Demand continuity in both languages suggests a stable investment horizon for developers and employers for the next 2–3 years.
- PostgreSQL's rise above MySQL in future demand is the most actionable signal from this survey. Database architects, DBA hiring managers, and curriculum designers should prioritize this shift immediately.
- The cloud-native stack — Docker, Kubernetes, AWS — has crossed into default territory. It is no longer a differentiator; it is the baseline expectation for production-grade engineering roles.
- The global developer community is young (25–34 dominant), highly educated (68% degree+), and concentrated in the US and India — patterns that directly shape open-source contributor pools, bootcamp markets, and hiring pipeline geography.

Next Steps

1. Build ML models to predict language adoption through 2026
2. Integrate GitHub API data to corroborate open-source trends
3. Extend analysis to include salary data for compensation benchmarking
4. Publish reproducible pipeline to GitHub as an open portfolio project

65,437
Developers Surveyed

14
Rubric Points Covered

12
Dashboard Charts

6
Key Ecosystem Shifts

References & Acknowledgements

Sources, tools, and supporting materials used throughout the project

Data Source	Stack Overflow (2023). Annual Developer Survey 2023. survey_data_updated.csv. IBM Skills Network. https://stackoverflow.com/survey
Course Reading	Haider, M. (2015). Getting Started with Data Science. IBM Press. Chapter 3, pp. 60–62 — The Report Structure and Have You Done Your Job as a Writer?
Visualization	Chart.js v4.4.0 (2023). Open-source JavaScript charting library. https://www.chartjs.org — used for all 12 interactive dashboard visualizations.
Methodology	IBM Data Science Professional Certificate — Capstone Project M10–M15. Jupyter Notebook analysis series using Python, pandas, and numpy.
Framework	IBM Data Science Findings Report Framework (video lecture). Cover → Executive Summary → Introduction → Methodology → Results → Discussion → Conclusion → Appendix.
Presentation	PptxGenJS v4.0.1 (2024). Node.js library for programmatic PowerPoint generation. https://gitbrent.github.io/PptxGenJS/

github.com/[your-username]/devpulse-2023 | Full Jupyter notebooks, interactive dashboard, and raw dataset available as open-source portfolio project